

MedRAG

Evidence-grounded clinical guideline question answering

From scattered clinical evidence to a traceable answer.

MedRAG retrieves relevant passages from indexed guidelines and PubMed abstracts, then uses an LLM to generate a concise answer with evidence, sources, and confidence.

Problem statement

Clinical guidance is extensive, frequently updated, and distributed across long-form documents and research sources. Finding the most relevant passage and explaining it consistently is slow, while an unconstrained language model can produce answers without a reliable evidence trail.

Context

MedRAG applies retrieval-augmented generation (RAG) to clinical guideline Q&A.; A domain-agnostic core handles indexing, retrieval, generation, evaluation, and service orchestration. A MedRAG plugin supplies clinical ingestion rules, metadata, prompts, and safety policy.

Objectives

- Retrieve clinically relevant evidence from approved indexed sources.
- Generate readable answers that expose evidence and source provenance.
- Provide consistent access through CLI, FastAPI, and Streamlit interfaces.
- Reduce unsafe behavior with input and output guardrails and repeatable evaluations.

01

Grounded answers

Uses retrieved guideline and research context instead of free-form generation.

02

Traceable evidence

Returns evidence snippets, sources, metadata, and a retrieval-derived confidence score.

03

Managed knowledge

Supports PDF upload, deletion, source status, and explicit collection rebuilds.

04

Safety boundary

Screens prompt injection and personalized or overconfident medical advice at the API edge.

OPERATING MODEL

How MedRAG works

The system separates knowledge preparation from live question answering. Indexing is explicit and repeatable; query-time work stays focused on safety checks, retrieval, generation, and structured response delivery.

Stage	Conventional search	MedRAG workflow	System output
Source preparation	Manually locate and scan documents	Parse PDFs and fetch configured PubMed abstracts	Normalized clinical documents
Knowledge organization	Rely on filenames or ad hoc notes	Chunk, enrich metadata, embed, and index in Qdrant	Searchable evidence collection
Question answering	Open documents and synthesize manually	Retrieve top-k context and generate a grounded answer	Answer, evidence, sources, confidence
Quality and safety	Depends on individual review practice	Run API guardrails and DeepEval quality metrics	Blocked unsafe output or assessed response

Key inputs

- Clinical guideline PDFs placed in or uploaded to the MedRAG guideline source directory.
- Configured PubMed search queries and retrieval limits.
- Runtime credentials for OpenAI and LlamaParse; Groq is optional for guardrails.
- Index configuration: embedding model, chunk size, overlap, query mode, and top-k.

Structured output

Answer	Evidence	Sources	Confidence
Guideline-level natural-language synthesis	Retrieved supporting snippets	Source file, title, organization, specialty, evidence type	Retrieval-derived score when available

No time-saving benchmark is claimed here: the repository defines the automation path, but does not contain a controlled productivity study.

SYSTEM DESIGN

Architecture

One shared RAG service coordinates a plugin-based core. The MedRAG project definition selects its clinical configuration and ingestor; the three interfaces reuse the same retrieval and generation path.



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|--------------------------|--|
| 1. Collect and parse | LlamaParse converts clinical guideline PDFs to structured text. The PubMed reader adds abstracts from configured searches. Optional bootstrap documents support demonstrations. |
| 2. Enrich and index | MedRAG labels each document with source organization, specialty, and evidence type. Sentence chunks are embedded locally with BAAI/bge-small-en-v1.5 and written to a named Qdrant collection. |
| 3. Retrieve and generate | A user question becomes a vector search. The selected evidence is inserted into the clinical system prompt, and OpenAI generates a concise educational response. |
| 4. Validate and return | At the API boundary, the answer is checked against the MedRAG safety policy. The API returns a typed response containing answer, evidence, source metadata, and confidence. |

IMPLEMENTATION

Core modules and responsibilities

Layer	Primary modules	Responsibility
Interfaces	src/cli.py src/api/main.py src/ui/app.py	Developer commands, typed HTTP endpoints, and an interactive Streamlit client.
Service	src/core/service.py	Creates a project-specific service, builds or loads the index, and coordinates queries.
RAG core	indexer.py retriever.py generator.py	Chunks and embeds documents, retrieves evidence, builds prompts, and parses structured answers.
Project plugin	projects/medrag/config.py projects/medrag/ingestor.py	Defines clinical prompts, metadata, collection settings, safety policy, and source ingestion.
Operations	source_manager.py evals.py docker-compose*.yaml	Manages source files, runs quality checks, and packages local or production services.

API surface

Knowledge	GET /sources, POST /sources/upload, DELETE /sources/{filename}, POST /sources/reindex
Question answering	POST /query
Safety diagnostics	GET /guardrails/status, POST /guardrails/test-input, POST /guardrails/test-output
Evaluation	POST /evals/medrag/run, GET /evals/medrag/latest
Operations	GET /health

Deployment topology

Docker Compose runs Qdrant, a one-shot indexer, FastAPI, and Streamlit. The indexer waits for Qdrant and prepares the collection; the API starts after indexing; the UI calls only the API. Production assets add Nginx, EC2 bootstrap scripts, CloudFormation infrastructure, and GitHub Actions workflows.

TRUST AND QUALITY

Safety, evaluation, and operational controls

Control	Implemented behavior	Important boundary
Input guardrail	Groq-hosted Prompt Guard estimates whether a query is malicious or attempts prompt injection.	Runs only on POST /query and dedicated guardrail tests; CLI and eval calls bypass it.
Output guardrail	A project policy flags direct dosage instructions, personalized diagnosis or treatment, and overstated certainty.	Policy classification is a safety layer, not clinical review or regulatory approval.
Failure behavior	Missing key, timeout, network failure, or malformed model output is logged and allowed through.	The current design fails open to preserve availability; production risk owners should explicitly accept or change this posture.
Evaluation	DeepEval measures faithfulness, answer relevancy, and contextual relevancy against a golden dataset.	Scores are model- and dataset-dependent; they do not establish patient outcome benefit.

Operational checklist

- Use only sources that the deployment is authorized to process and retain.
- Keep the source collection current, then rebuild the index after source changes.
- Protect API keys and never commit populated environment files.
- Review retrieved evidence and answers for clinical appropriateness before relying on them.
- Run automated tests and the golden-dataset evaluation after prompt, model, embedding, or retrieval changes.
- Monitor guardrail availability, response failures, latency, and source freshness in production.

MedRAG should be positioned as an educational evidence assistant. It does not diagnose, prescribe, replace a clinician, or guarantee that indexed guidance is complete or current.

PRODUCT EXPERIENCE

Product walkthrough

The Streamlit application is a thin HTTP client over the FastAPI service. It centralizes knowledge management, question answering, evaluation results, and guardrail diagnostics without embedding retrieval logic in the UI.

MedRAG

Clinical evidence assistant

Ask

Sources

Evaluations

Guardrails

Ask a clinical guideline question

Answers are grounded in indexed evidence and are educational only.

What do guidelines say about first-line therapy for hypertension?

ASK

Evidence-grounded answer

Guidelines generally recommend lifestyle changes and selected first-line antihypertensive classes, individualized by comorbidities and clinical context. A qualified clinician should interpret targets and treatment choices.

CONFIDENCE 0.84

Evidence and sources

WHO_BP.pdf

guideline | cardiology

PubMed abstract

research_abstract | cardiology

Typical user flow

1	Prepare	Upload approved guideline PDFs and inspect PubMed ingestion settings.
2	Index	Rebuild the Qdrant collection so new or removed sources affect retrieval.
3	Ask	Submit a clinical guideline question through the UI or POST /query.
4	Review	Read the answer together with evidence snippets, sources, and confidence.
5	Assess	Run the evaluation suite and guardrail diagnostics after meaningful system changes.

Project status and next priorities

- Current scope: RAG core, MedRAG plugin, source management, typed API, Streamlit UI, evaluations, Docker/AWS deployment assets, and feature-branch I/O guardrails.
- Next: set product quality thresholds, expand the clinician-reviewed golden dataset, and choose a monitored production failure policy for guardrails.
- Before sensitive use: add authentication, authorization, audit logging, retention rules, and formal privacy, security, and clinical review.

Prepared from the MedRAG repository implementation and the supplied report's presentation pattern. Repository state reviewed 23 August 2026.